POC SUMMARY · BRIEF 12 · CLEAN ENERGY SOLUTIONS · V2

Relocatable clean power for a hospital on the move

Team Secure AI Grid · Pavan Manjunath, Thomas Prüfer | **Worked example** National Orthopaedic Hospital Enugu

THE PROBLEM: MARKET EVIDENCE, THEN THE WORKED EXAMPLE

About half of the hospitals in sub-Saharan Africa have reliable electricity [9], [24]. Most firms in sub-Saharan Africa report power outages (World Bank Enterprise Surveys) [26]. Africa holds about 60 % of the world's best solar resources but about 1 % of installed solar capacity [25]. In Nigeria, diesel averaged ₦3,277 a litre in May 2026, up 86 % in a year (NBS national average, not an Enugu pump price) [4]; federal teaching hospitals have seen electricity bills rise three to four times since the 2024 Band A tariff increase [3]; the country plans to power 3,700 health centres with renewables [7] and has lost about 15,000–16,000 doctors to emigration in five years [6].

The worked example is the National Orthopaedic Hospital Enugu, 350 beds (brief). Reported by the hospital's head engineer in Problem Brief 12: about 6 h of grid a day, about 1,550 kW connected, sensitive equipment damaged by unstable supply, and a theatre that waits 10–30 s for the generator at every grid drop [1], [11]. Published: Band A tariff ₦160/kWh and feeder hours [5]. The brief's scenario: the hospital moves to a new building in about five years, and federal asset rules prevent it from disposing of solar fixed to today's buildings, so rooftop solar would be stranded [11]. Modelled from open data: grid class D, "Poor", 9 substations within 50 km; 5.48 peak-sun hours a day confirmed by four open sources [15]–[18]. Precedents: Dala, Kano (1.5 MW solar hybrid) [2]; EU–Enugu Solar for Health (₦4.3 bn) [10]; a Tanzanian camp hospital on containerised solar, built to move [8].

WHAT WE ARE SOLVING FOR: THE THREE A'S

ACCESS	AFFORDABILITY	AVAILABILITY
Today: about 6 h of grid a day against 1,550 kW of need [1]; the state publishes feeder supply hours [5]. **We help (modelled):** any site read from open data in seconds; for Enugu 9 substations within 50 km, grid class D, 5.48 peak-sun hours, and the 303 kW critical load first [13].	**Today:** diesel at ₦3,277/L, up 86 % in a year [4]; grid at ₦160/kWh [5]; bills three to four times higher since 2024 [3]. **We help (modelled):** ESCO service: ₦0 upfront, ₦69.1 M a month in place of ₦299.2 M of modelled diesel. Bought outright (₦3.85 bn): repaid by diesel savings, with the move, in 1.6 years [13].	**Today:** the grid drops without warning and switchovers break equipment [1]; the hospital moves in about five years [11]. **We help (modelled):** 4 of 4 stress tests pass; 100 % of the critical load in every modelled outage up to 24 h, whatever hour it starts; 88 % of the investment moves with the hospital [13].

GridRescue AI is vendor-neutral decision-support software: pin a hospital on the map and it reads the site from open data, simulates every outage hour by hour, designs the solar + energy system + UPS + generator stack, prices it, plans the move and writes the ESG report, in minutes.

WHAT WAS BUILT DURING THE GLOBAL ENERGY HACKATHON 2026

What we built, in short: a front-end GUI grid mapper (sun · wind · water · grid lenses and the new-site lens) → a basic ML classification of the green energy system (Random Forest picks the energy system; the hourly simulation verifies it) → a location-based ESG report, downloadable for any site.

MODULE	WHAT IT DOES (WHERE IN THE APP)
1 Grid mapper	Sun, wind, water and grid lenses from open data (NASA POWER, PVGIS, PVWatts, Global Solar Atlas [15]–[18]; OpenStreetMap [19]); substations within 50 and 100 km; published supply hours and tariff; the "hospital is moving" lens ranks areas within 12 km on grid access and distance (Explorer).
2 Relocation plan	The brief's five starter questions [11]: two sites, theatres-first load list, movable energy system, energy type, no-sun case, move plan, money (5- and 10-year cost model, ownership, funders), impact, data sources (Plan).
3 3D outage simulation and ML	Hour by hour, four stress tests and the outage-survival curve in 3D on the hospital's own site; ML classification of the energy system (Random Forest, 30 trees, 80.4 % agreement with the engine on held-out synthetic sites; the app shows the pick with a 76 % vote share), verified by the hourly simulation (Simulate, Advisor).
4 ESG report	Scope 1 and Scope 2 under the GHG Protocol [14], E · S · G lines, carbon value at the buyer's price only after verification; saves as PDF for the selected site (Impact, Simulate, Report).

ESG report download: Relocation plan › Impact tab (Download ESG report), Simulate page (ESG report button) and Report page; it works for any pinned location (tested: Enugu, Kano, Nairobi, Accra) and gives Scope 1 · Scope 2 (GHG Protocol), E·S·G report lines and carbon value at the buyer's price — a board-ready, business-standard report (modelled, ±30 %).

Also built: offline mode, one-click start, tests and CI, research-repository files (code Apache-2.0, OpenStreetMap data ODbL). Pre-existing before the sprint: the web-application frame and 3D scene kit (TO CONFIRM with Pavan before the disclosure statement is published). AI tools: an AI coding assistant for code, documents and the video pipeline (wording TO CONFIRM).

Three layers of protection: (1) online UPS on theatre and ICU circuits, 0 ms switchover (modelled); (2) solar by day, energy containers by night; (3) the existing generator last. **The engine** runs hour by hour per outage case; an outage started at each of the 24 hours must be carried for up to 24 h (survival check in the style of NREL's REopt); every step is shown with its source.

FIVE OUTAGE CHALLENGES THE PLATFORM ADDRESSES

RISK CATEGORY (NOT A RANKING)	BUILT	EXTENSION / PLANNED
1 Grid supply interruptions and load shedding	The 12 h evening outage (design outage); survival check at every start hour.	Feeder or substation trips; scheduled band cuts.
2 Transmission and distribution failures	The 24 h national grid collapse case.	Frequency collapse; outage groups by feeder, district or state market.
3 Severe weather and infrastructure damage	The worst solar month case.	Storm, flood and heat-wave groups.
4 Diesel shortages, cost and generator failures	Diesel modelled as the last tier; the 18 h outage case.	—
5 Cyber or operational disruptions	Not implemented.	Loss of grid, communication or control as a further stress scenario; never cyber detection or prevention.

WORKED EXAMPLE: THE BRIEF'S STARTER QUESTIONS

STARTER QUESTION [11]	ENGINE ANSWER FOR ENUGU (MODELLED, ±30 %) [13]
Which loads must never lose power?	Critical load 303 kW (theatres 91 kW) on a 500 kVA online UPS, 0 ms switchover (modelled); protected load 641 kW; daily use about 14,880 kWh.
What should be installed?	1,670 kWp ground-mounted solar; a 5,020 kWh containerised energy system (8.7 h of protected load) in 3 energy containers (5 containers in all, with inverter skids and the UPS e-house); 500 kVA online UPS; the existing 1,000 kVA generator last. Capex ₦3.85 bn.
Will the theatres stay on?	4 of 4 stress tests pass (12 h evening outage, 24 h national grid collapse, worst solar month, 18 h outage); 100 % of the critical load in every modelled outage up to 24 h, whatever hour it starts.
Can diesel savings repay it, and how fast?	Bought outright: diesel savings repay the system and the year-5 move in 1.6 years. ESCO service: ₦0 upfront, ₦69.1 M a month vs ₦299.2 M of modelled diesel; saves from month one if the real bill (still owed) exceeds ₦117.9 M a month.
How does it move in five years?	To lens area #1, 4 km north (grid-access score 98/100, substation 1.3 km away), in about three weeks for ₦678 M; 88 % of the investment reused; theatres on UPS and diesel meanwhile.
Who owns it, and who funds it?	An ESCO owns, insures and moves the kit, with concessional debt. Funders: REA public-sector solarisation (Dala, Kano [2]); EU–Enugu Solar for Health [10]; development-bank financing for health-facility electrification [7].
Which energy system?	Without a generator: containerised energy system ₦4.65 bn versus hydrogen ₦8.18 bn (almost twice; about 35 % round trip; cannot move). Grid-first fallback when an energy system is not yet possible: UPS on the theatres, the generator for outage hours, the energy system later.

ALL ENERGY COSTS OF THE PROTECTED LOAD (MODELLED) [13]	5 YEARS	10 YEARS
Today: grid for 6 h + modelled diesel	₦18.8 bn	₦37.6 bn
Grid only (reference: a 24 h grid, which the hospital does not have [1])	₦3.5 bn	₦7.0 bn
Solar service: fee + remaining diesel + remaining grid	**₦7.9 bn**	**₦15.9 bn**

Saved versus diesel: ₦10.9 bn over 5 years, ₦21.8 bn over 10 years (rounded). Bought-outright view on the Simulate page: payback 1.3 years, US$ 31.83 M saved over 15 years.

BEFORE → AFTER (ENGINE, FOR THE MODELLED OUTAGES)

KPI FOR THE PROTECTED LOAD	TODAY (MODELLED)	WITH THE SYSTEM (MODELLED)
Theatre-hours a year without grid (12,480 a year, 4 suites)	4,992 (40 %)	0
Generator switchovers reaching equipment	about 1,460 a year	0 (online UPS)
Diesel	1,042,790 L a year	176,112 L a year (866,678 L, 83 %, avoided)
CO_2	2,795 t a year	549 t a year (2,246 t net avoided: 2,323 t Scope 1 − 77 t Scope 2)
All energy costs, year one	₦3,764.5 M (grid + modelled diesel)	₦1,588.9 M (fee + remaining diesel and grid)
Sun captured by the planned solar field	0	about 6.2 MWh a day captured, 5.9 MWh used by the system

±30 %, to be confirmed with the hospital's load data. The ₦69.1 M a month service fee is the hospital's example energy-as-a-service cost, not GridRescue AI's revenue.

THE ESG REPORT, BEFORE ANYTHING IS BOUGHT

Nigeria's FRC is rolling out IFRS S1 and S2, which ask for Scope 1 and Scope 2 emissions [20], and calls sustainability governance a board duty [21]; hospitals use about 2.5 times the energy of a commercial building [22] and already certify green [23]. Enugu (modelled): 2,246 t CO_2 a year net = 2,323 t Scope 1 (2.68 kg CO_2/L, IPCC default) − 77 t Scope 2 (grid power to charge the energy system, national grid factor) [14]; about 22,460 t over 10 years [13].

WHO IS THE USER AND WHO PAYS

User: the facilities and engineering team. **Payer:** the hospital or health network buys the software; an ESCO finances, owns and moves the hardware. Three money stories, never mixed:

(A) THE HOSPITAL'S SAVINGS	(B) THE ESCO'S PROJECT CASH FLOW	(C) GRIDRESCUE AI'S SOFTWARE REVENUE
Diesel avoided minus service costs. Modelled, all energy costs of the protected load: ₦37.6 bn → ₦15.9 bn over 10 years. Simple payback applies only when the hospital buys outright, not to the investor.	The ₦69.1 M a month service fee (the hospital's cost, not our revenue), capex ₦3.85 bn, O&M, the relocation clause, concessional debt; funders such as the REA programme, Solar for Health, development banks.	Assessment US$ 2,000 per site; licence US$ 3,000 per site per year; enterprise and white-label fees; later a performance fee (proposed prices). Illustrative: 10 sites → US$ 50,000 in year one; 20 → US$ 100,000.

BUSINESS MODEL

Business Model Canvas (hybrid: assessment + licence + enterprise partnerships; the full nine blocks are in the canvas deliverable). **Customer segments:** hospitals and clinics on weak grids (first: federal and teaching hospitals, Nigeria) · healthcare networks and health programmes · ESCOs, energy-system integrators, hospital energy contractors · utilities (feeder planning) · later: data centres, water, telecom, industrial and agricultural sites. **Value proposition:** one workflow from a map pin to an investment-ready plan · relocatable designs · minutes instead of weeks · transparent assumptions, no hardware sales. **Channels:** direct institutional sales · ESCO and integrator partners (white-label) · the free grid mapper as the front door.

MARKET RESEARCH · TOOLS, WHAT IS DIFFERENT, WHAT SCALES

Tool	Grid mapping	Hourly outage sim.	ML energy-system pick	Relocation plan	Financing / cost	ESG report	No expert needed
⚡ REopt (NREL)	✗	✓	✗	✗	◐	✗	✗
⚙ HOMER Pro (UL Solutions)	✗	✓	✗	✗	✓	✗	✗
✚ HOMER Powering Health (World Bank/ESMAP)	✗	✗	✗	✗	✓	✗	–
$ Odyssey Energy Solutions	✗	✗	✗	✗	✓	✗	–
⌖ MapYourGrid / OpenStreetMap	✓	✗	✗	✗	✗	✗	–
★ GridRescue AI	✓	✓	✓	✓	✓	✓	✓

✓ *yes* ◐ *partly* ✗ *no* – *not stated in our sources* *Sources: NREL, UL Solutions, World Bank/ESMAP, Odyssey and MapYourGrid websites [27]–[31].*

What scales · any pin on the map · a new country adds its tariff table and local prices · from hospitals to every critical facility (data centres, water, telecom) · further outage types (weather, transmission, cyber / operational as stress scenarios).

Positioning. HOMER Powering Health Tool (World Bank/ESMAP, free): a sizing and cost tool, not an outage-by-outage simulation; no relocation, no grid mapping, no ESG report [29]. REopt [27] and HOMER Pro [28] are built for energy experts; Odyssey serves developers [30]; MapYourGrid is our data partner [31].

What none of them does in one flow, and GridRescue AI does today: (1) from a map pin to an investment-ready hospital plan in minutes, on open data, with no API keys and no expert; (2) the grid mapper with the "hospital is moving" new-site lens: substations and lines within 50 and 100 km, published supply hours, areas ranked for the new building, then a relocation plan with a move cost and a reuse percentage; (3) every energy-system family simulated hour by hour on the site's own sun, grid hours and load, four stress tests, with a vendor-neutral ML classification of the energy system that the simulation verifies; (4) local market assumptions built in: the NBS diesel price, the state tariff, Nigerian funders and programmes; (5) the ESG report (Scope 1 and Scope 2, E·S·G lines) and the before → after KPIs written before anything is bought. The simulation itself is not new; the integrated, hospital-specific, relocatable, financeable workflow is. **Why pay when REopt is free:** the hospital workflow, relocation planning, locally validated data, financing-ready outputs, and an answer in minutes without an expert.

THE INVESTOR ASK

The ask. A proposed seed investment of US$ 150,000 in two milestone-linked tranches of US$ 75,000, the second after the simulation engine passes agreed validation tests against hospital data and a pilot partner is secured; lean pilot option US$ 75,000, about 6 months. It funds the software business, not hospital hardware such as solar, energy systems or generators (those need separate project financing, e.g. an ESCO with concessional debt).

USE OF FUNDS	US$	USE OF FUNDS	US$
Software engineering and productionisation	70,000	Hospital pilot integration and field validation	15,000
Energy engineering and model validation	20,000	Customer acquisition and partner onboarding	10,000
Security, testing and hardening	12,000	Legal, contracts and IP	5,000
Cloud hosting, weather and geospatial data	8,000	Contingency	10,000
Total	**150,000**	**Two milestone-linked tranches**	**2 × 75,000**

Planning estimates, not quotations or commitments. Milestones (targets, not achievements): commercial release in 9–12 months · 2 paid pilot sites by month 9 · 10 paying sites by month 12.

ROADMAP, 6–18 MONTHS

MONTHS	WHAT HAPPENS
0–3	Validate the engine with the hospital's load audit and generator logs; security hardening.
3–9	Two paid pilots (Enugu first) with an ESCO partner and the local distribution company.

MONTHS	WHAT HAPPENS
9–12	Commercial release; 10 paying sites.
12–18	Utilities and other critical facilities (e.g. clinic cold chain, Brief 01 [12]); further outage scenarios (weather groups, cyber/operational contingencies); new countries by adding a tariff table.

HOW WE MEET THE JUDGING CRITERIA [11]

JUDGED ON	OUR EVIDENCE
Impact · 25 %	Theatre-hours without grid 4,992 (40 %) → 0; 866,678 L of diesel (83 %) and 2,246 t CO_2 (net) avoided a year; about 1,460 generator switchovers a year removed by the online UPS (all modelled) [13].
Innovation · 25 %	Not the simulation itself but the integrated workflow: site read → outage simulation → vendor-neutral design → cost and payback → ESG report, with the "hospital is moving" lens.
Sustainability · 25 %	Containers and ground mounts move (~~₦~~678 M, 88 % reused); about 22,460 t CO_2 and ~~₦~~21.8 bn saved over 10 years (modelled).
Feasibility · 15 %	Published hours, tariff and four solar sources; 4/4 stress tests (modelled); hospital-data validation first.
Scalability · 10 %	Global open data; a new country needs only its tariff table; vendor-neutral, we sell no hardware.

HOW FIGURES ARE LABELLED Reported = the hospital's head engineer or the brief; published = the cited source; modelled = a GridRescue AI engine result (±30 %, to be confirmed with the hospital's load data). "0 ms switchover" is the modelled behaviour of the online UPS; "100 % of the critical load" holds for the modelled outages, not as a universal guarantee.

OUR TEAM SUBMITS A demo video (under 5 minutes), a pitch video (under 5 minutes), the 5-slide deck, the one-pager, this POC summary, the Business Model Canvas, the financial model and the disclosure statement.

GridRescue AI turns energy planning into investment-ready decisions. We start with hospitals, where power is life, and grow to every critical facility. Map the grid. Keep the theatres on. Move with the hospital.

REFERENCES

[1] National Orthopaedic Hospital Enugu, head engineer, "Problem statement and background shared with the Global Energy Hackathon organizers," 2026 (reproduced in Problem Brief 12).
[2] Gazette Nigeria, "FG inaugurates 1.5MW solar hybrid grid at Dala Orthopedic Hospital," 2025. [Online]. Available: https://gazettengr.com/fg-inaugurates-1-5mw-solar-hybrid-grid-at-dala-orthopedic-hospital/
[3] AllAfrica, "Nigeria: Electricity Tariff... Life-Saving Services Threatened in Teaching Hospitals," 2024. [Online]. Available: https://allafrica.com/stories/202406100036.html
[4] Legit.ng, "NBS: Diesel Price Rises 86.40% To N3,277/Litre In May 2026," 2026. [Online]. Available: https://www.legit.ng/business-economy/energy/1716140-nbs-diesel-price-rises-8640-n3277litre-2026/
[5] Nairametrics, "Enugu slashes electricity tariff for Band A customers to N160/kWh effective August 1," 2025. [Online]. Available: https://nairametrics.com/2025/07/21/enugu-slashes-electricity-tariff-for-band-a-customers-to-n160-kwh-effective-august-1/
[6] Techeconomy, "Japa: 16,000 doctors left Nigeria in five years – Minister," 2024. [Online]. Available: https://techeconomy.ng/japa-16000-doctors-left-nigeria-in-five-years-minister
[7] BusinessDay, "Nigeria to electrify 3,700 healthcare centres with renewables," 2024. [Online]. Available: https://businessday.ng/energy/power/article/nigeria-to-electrify-3700-healthcare-centres-with-renewables/
[8] Africa Renewable Energy (ARE) member case study, "Nduta Main Hospital solar system, Tanzania." [Online]. Available: https://renewelec.org/case-study/nduta-main-hospital-solar-system/
[9] World Health Organization, "Electricity in health-care facilities," Fact sheet. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/electricity-in-health-care-facilities
[10] ThisDay, "EU, Enugu Govt to Spend N4.3bn on Solar Energy in Healthcare Centres," 2024. [Online]. Available: https://www.thisdaylive.com/2024/11/20/eu-enugu-govt-to-spend-n4-3bn-on-solar-energy-in-healthcare-centres/
[11] Global Energy Hackathon, "Problem Brief 12: Relocatable solar for a hospital on the move," Clean Energy Solutions track, hackathon.anneru.com (success metrics, constraints and starter questions).
[12] Global Energy Hackathon, "Problem Brief 01" (clinic cold chain), New Technologies & Hardware track, hackathon.anneru.com.
[13] GridRescue AI, "GridRescue engine, Enugu worked example," output of the running proof-of-concept application for the National Orthopaedic Hospital Enugu site (modelled, ±30 %; to be confirmed with the hospital's load data).
[14] World Resources Institute and WBCSD, The Greenhouse Gas Protocol: A Corporate Accounting and Reporting Standard, rev. ed. [Online]. Available: https://ghgprotocol.org/corporate-standard
[15] NASA Langley Research Center, "POWER Data Access Viewer" (solar and meteorological data). [Online]. Available: https://power.larc.nasa.gov/
[16] European Commission Joint Research Centre, "PVGIS: Photovoltaic Geographical Information System." [Online]. Available: https://re.jrc.ec.europa.eu/pvg_tools/en/
[17] National Renewable Energy Laboratory, "PVWatts Calculator." [Online]. Available: https://pvwatts.nrel.gov/
[18] World Bank Group and Solargis, "Global Solar Atlas." [Online]. Available: https://globalsolaratlas.info/
[19] OpenStreetMap contributors, "OpenStreetMap" (substations, power lines and roads). [Online]. Available: https://www.openstreetmap.org/
[20] Financial Reporting Council of Nigeria, "Roadmap Report for the Adoption of IFRS Sustainability Disclosure Standards in Nigeria (Amended 2026)," 2026. [Online]. Available: https://frcnigeria.gov.ng/wp-content/uploads/2026/03/Roadmap-Report-for-the-Adoption-of-IFRS-Sustainability-Disclosure-Standards-in-Nigeria-Amended-2026-.pdf
[21] Financial Reporting Council of Nigeria, "Climate and sustainability governance now a core board duty," 2025. [Online]. Available: https://frcnigeria.gov.ng/2025/11/28/climate-and-sustainability-governance-now-a-core-board-duty-frc-tells-directors/
[22] International Finance Corporation, "EDGE (Excellence in Design for Greater Efficiencies)" green building certification programme, guidance for hospitals. [Online]. Available: https://edgebuildings.com/
[23] ThisDay, "Evercare Hospital Lekki Becomes First EDGE-Certified Hospital in Nigeria," 2022. [Online]. Available: https://www.thisdaylive.com/2022/05/27/evercare-hospital-lekki-becomes-first-edge-certified-hospital-in-nigeria/
[24] World Health Organization, World Bank, IRENA and SEforALL, Energizing health: accelerating electricity access in health-care facilities. Geneva: WHO, 2023. [Online]. Available: https://www.who.int/publications/i/item/9789240066960
[25] International Energy Agency, Africa Energy Outlook 2022. Paris: IEA, 2022. [Online]. Available: https://www.iea.org/reports/africa-energy-outlook-2022
[26] World Bank, "Enterprise Surveys: infrastructure indicators (power outages)." [Online]. Available: https://www.enterprisesurveys.org/
[27] National Renewable Energy Laboratory, "REopt: Renewable Energy Integration and Optimization." [Online]. Available: https://reopt.nrel.gov/
[28] UL Solutions, "HOMER Pro microgrid software." [Online]. Available: https://www.homerenergy.com/products/pro/index.html
[29] World Bank / ESMAP and HOMER Energy, "HOMER Powering Health Tool." [Online]. Available: https://poweringhealth.homerenergy.com/
[30] Odyssey Energy Solutions, "Odyssey platform." [Online]. Available: https://www.odysseyenergysolutions.com/
[31] MapYourGrid, "Mapping the world's power grid in OpenStreetMap." [Online]. Available: https://mapyourgrid.org/